

BIG DATA ANALYTICS

Module 1

Big Data

Data is defined as the quantities, characters, or symbols on which operations are performed by a computer.

Data may be stored and transmitted in the form of electrical signals and recorded on magnetic, optical, or mechanical recording media.

Big Data is also data but with a huge size.

In short such data is so large and complex that none of the traditional data management tools are able to store it or process it efficiently.

Examples:

1. Stock Exchange generates about one terabyte of new trade data per day.
2. Social media site Facebook data

Big Data includes huge volume, high velocity and extensible variety of data.

The data in it may be structured data, Semi Structured data or unstructured data.

Big data also involves various tools, techniques and frameworks.



BIG DATA DEFINITION..

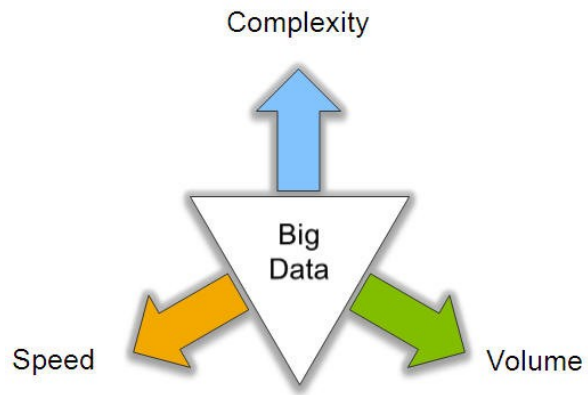
No single standard definition...

“**Big Data**” is data whose scale, diversity, and complexity require new architecture, techniques, algorithms, and analytics to manage it and extract value and hidden knowledge from it...

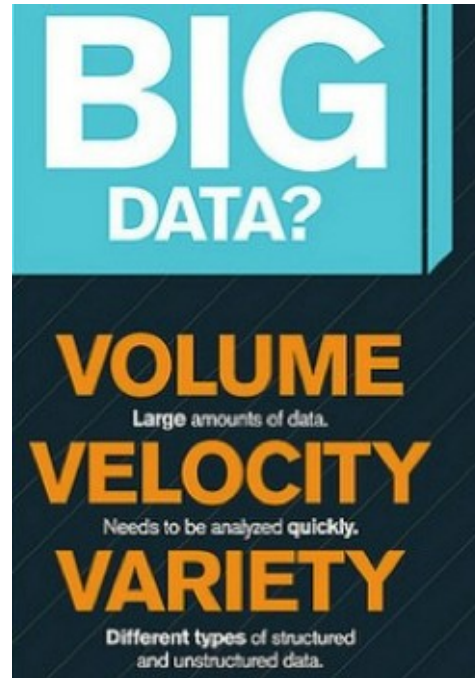
- “Big Data refers to vast amounts of data that cannot be processed effectively with traditional data processing techniques”

Traditional VS Big Data

Sl. No.	Traditional Data	Big Data
1.	Here the data is “Structured” data	Here the data is “Unstructured or Semi structured” data
2.	The size of the data is very small	The size is more than the traditional data size
3.	Here the data is Centralized	Here the data are distributed
4.	It is easy to work or manipulate	It is difficult to handle the data
5.	Normal system configuration is sufficient to process	High system configuration is required to process the data
6.	A traditional database tools is enough	Special kind of tools are required
7.	Normal functions are enough to manipulate the data	Requires special kind of functions to manipulate the data



3 V'S OF BIG DATA ANALYTICS

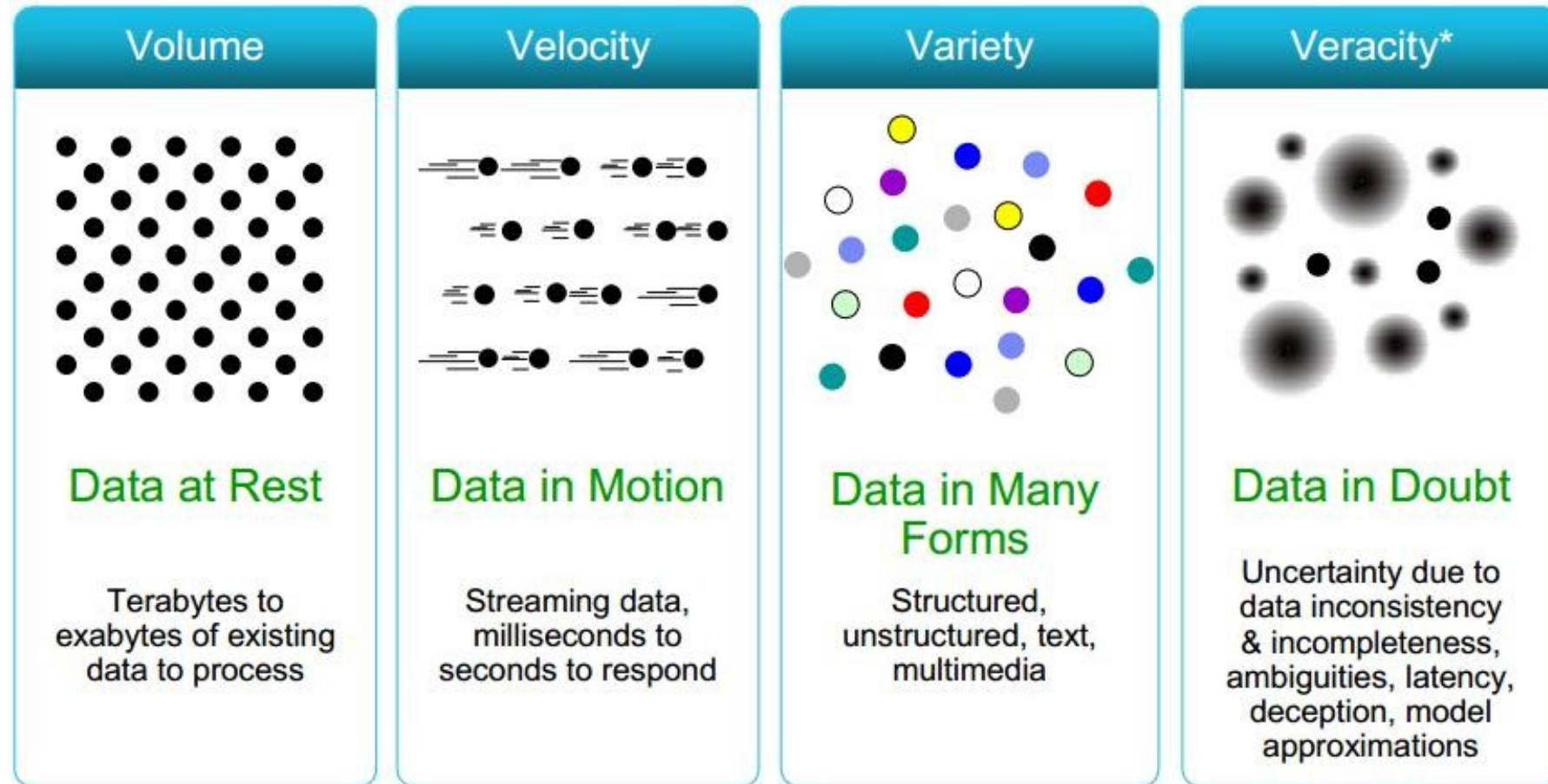


➤ **Variety**

➤ **Velocity**

➤ **Volume**

SOME MAKE IT 4V'S



1. Volume

- The name Big Data itself is related to a size which is enormous.
- Size of data plays a very crucial role in determining value out of data.
- Also, whether a particular data can actually be considered as a Big Data or not, is dependent upon the volume of data.
- Hence, 'Volume' is one characteristic which needs to be considered while dealing with Big Data.
- This refers to the data that is tremendously large.
- As shown in figure below, the volume of data is rising exponentially.
- In 2016, the data created was only 8 ZB and it is expected that, by 2020, the data would rise up to 40 ZB, which is extremely large.

1. Volume

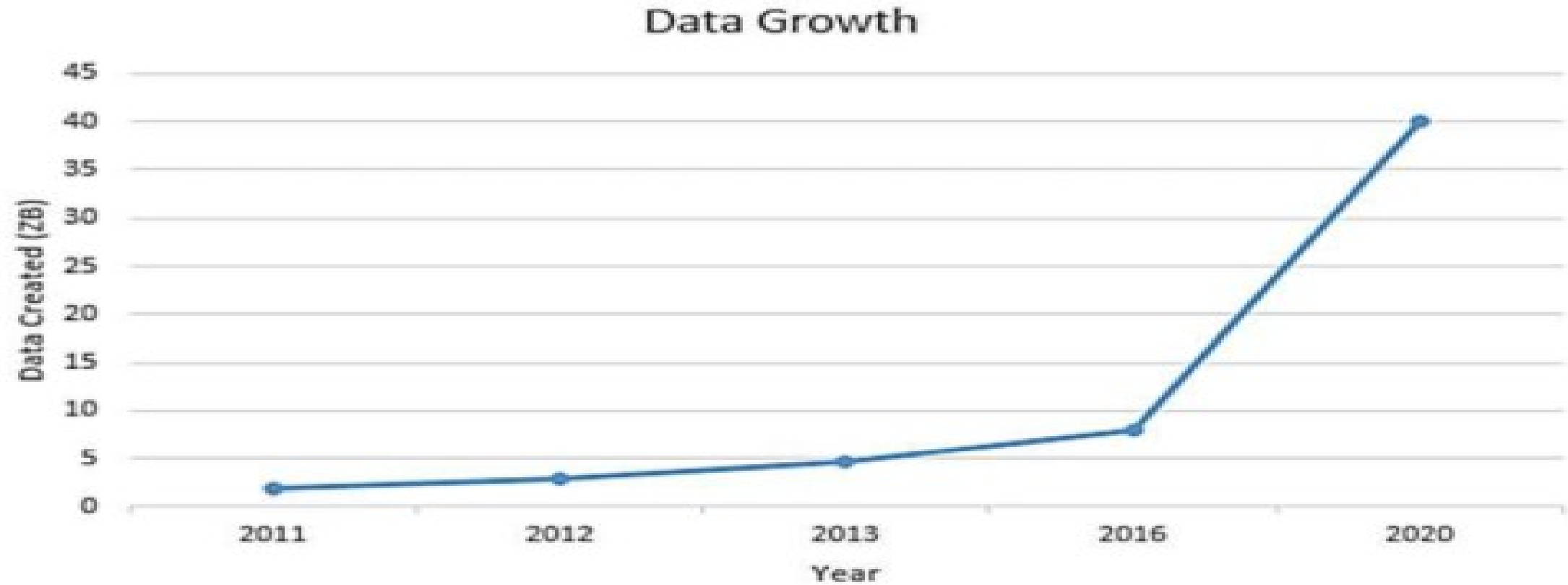


Figure 1.3: Big Data Volume

Units used to measure data storage

➤ **Bit (b):**

➤ **Byte (B):**

- 1 Byte = 8 bits.
- Basic unit used for measuring small amounts of data, such as a single character.

➤ **Kilobyte (KB):**

- 1 KB = 1,024 bytes.
- Often used to quantify the size of small files or documents.

➤ **Megabyte (MB):**

- 1 MB = 1,024 KB = 1,048,576 bytes.
- Commonly used to describe the size of files like high-resolution photos or short videos.

➤ **Gigabyte (GB):**

- 1 GB = 1,024 MB = 1,073,741,824 bytes.
- Frequently used to measure the storage capacity of devices like hard drives and the size of large files such as movies or software applications.

➤ **Terabyte (TB):**

- 1 TB = 1,024 GB = 1,099,511,627,776 bytes.
- Used for large-scale data storage, such as in enterprise servers or data centers.

➤ **Petabyte (PB):**

- 1 PB = 1,024 TB = 1,125,899,906,842,624 bytes.
- Used to measure extensive data storage capacities, often found in big data analytics, scientific research, or large-scale digital libraries.

➤ **Exabyte (EB):**

- 1 EB = 1,024 PB = 1,152,921,504,606,846,976 bytes.
- Commonly used in describing global annual internet traffic or large-scale cloud storage.

➤ **Zettabyte (ZB):**

- 1 ZB = 1,024 EB = 1,180,591,620,717,411,303,424 bytes.
- Used in scenarios involving vast amounts of data, such as global data centers or data generated by IoT devices.

➤ **Yottabyte (YB):**

- 1 YB = 1,024 ZB = 1,208,925,819,614,629,174,706,176 bytes.
- Represents an enormous amount of data, typically used in futuristic scenarios or theoretical discussions about data storage capacities.

2. Velocity

- The term velocity refers to the speed of generation of data.
- Big Data Velocity deals with the speed at which data flows in from sources like business processes, application logs, networks, and social media sites, sensors, Mobile devices, etc.
- The flow of data is massive and continuous.
- The speed of data accumulation also plays a role in determining whether the data is categorized into big data or normal data.

2. Velocity

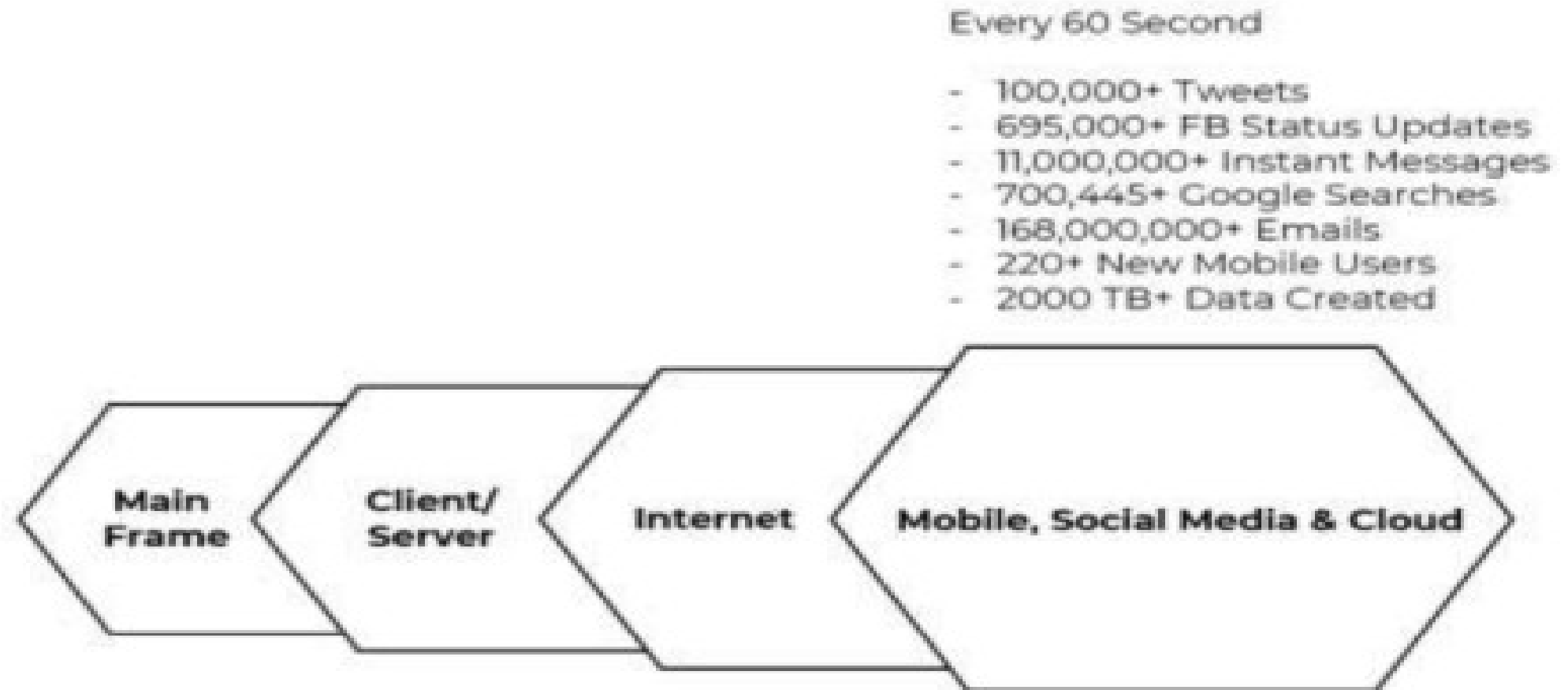
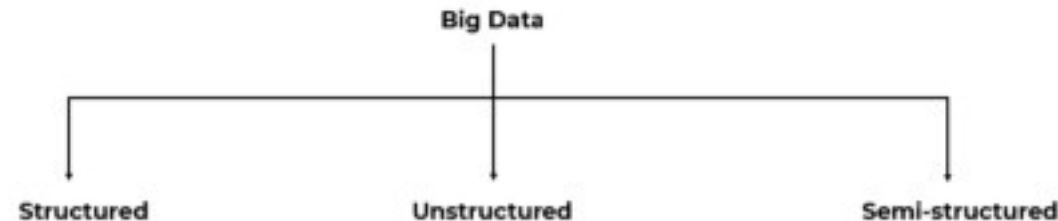


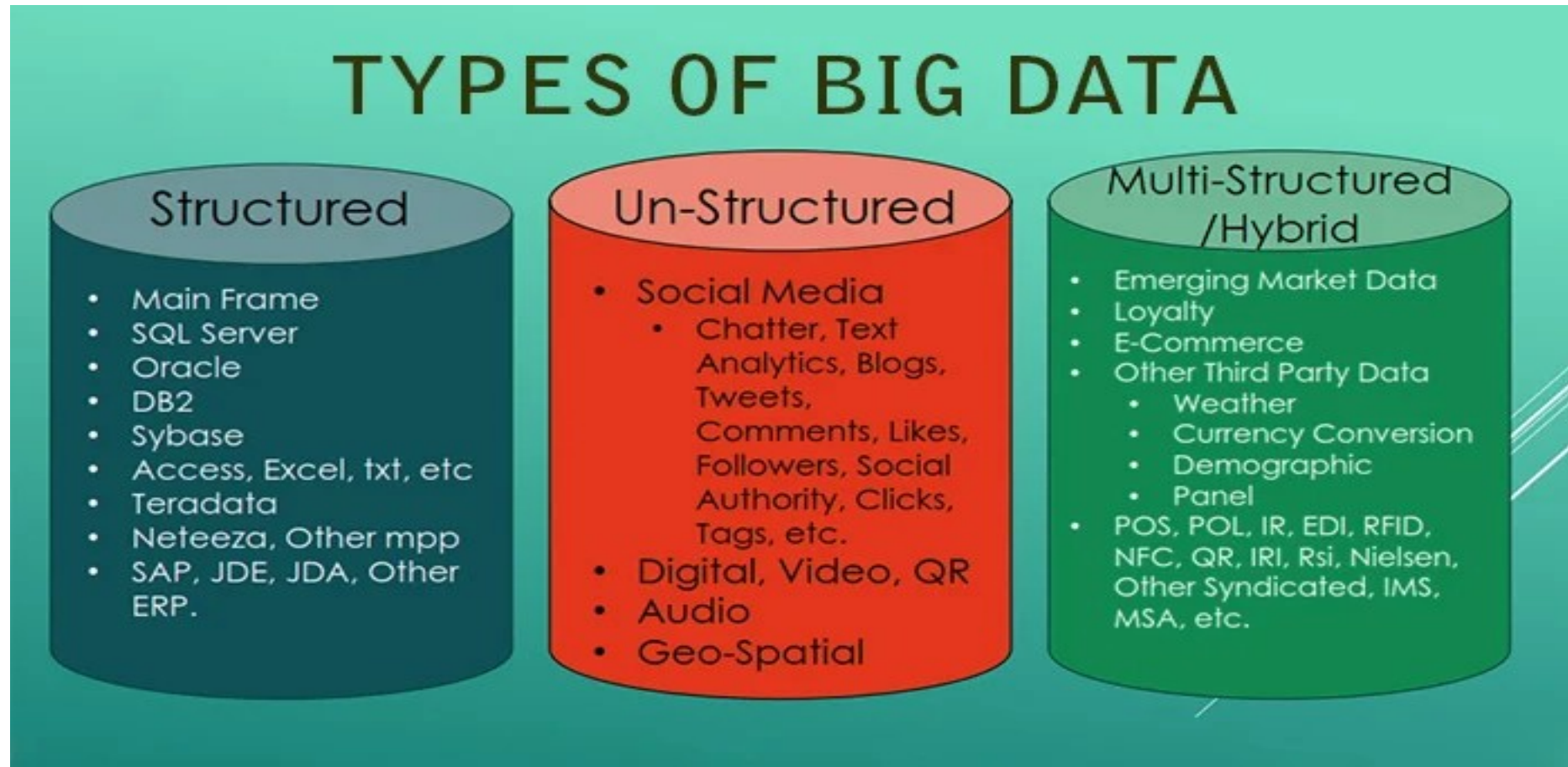
Figure 1.2: Big Data Velocity

3. variety

- Variety of Big Data refers to structured, unstructured, and semi structured data that is gathered from multiple sources.
- The type and nature of data is having great variety.
- During earlier days, spreadsheets and databases were the only sources of data considered by most of the applications.
- Nowadays, data in the form of emails, photos, videos, monitoring devices, PDFs, audio, etc. are also being considered in the analysis applications.



3. Variety



Structured

- Any data that can be stored, accessed and processed in the form of fixed format is termed as a structured Data.
- 20% of the total existing data and is used the most in programming and computer-related activities.
- There are two sources of structured data - machines and humans.
- All the data received from sensors, weblogs, and financial systems are classified under machine-generated data.
- These include medical devices, GPS data, data of usage statistics captured by servers and applications.
- Human-generated structured data mainly includes all the data a human input into a computer, such as his name and other personal details.
- When a person clicks a link on the internet, or even makes a move in a game, data is created

- Tools generates structured data:
 - Data Marts
 - RDBMS
 - Greenplum
 - TeraData
- Example: An 'Employee' table in a database is an example of Structured Data.

EmpId	ManagerId	Name	Department	Salary	City
1	1	Alex Smith	Admin	\$90,000	Boulder
2	1	Amy Mars	Admin	\$50,000	Longmont
3	1	Logan Mars	Admin	\$70,000	Longmont
4	1	James Mont	Marketing	\$55,000	
5	6	John Smith	Marketing	\$60,000	Boulder
6	6	Lily Mars	Marketing	\$95,000	

Unstructured

- Any data with unknown form or the structure is classified as unstructured data.
- The rest of the data created, about 80% of the total account for unstructured big data.
- Unstructured data is also classified based on its source, into machine-generated or human-generated.
- Machine-generated data accounts for all the satellite images, the scientific data from various experiments and radar data captured by various facets of technology.
- Human-generated unstructured data is found in abundance across the internet since it includes social media data, mobile data, and website content.
- This means that the pictures we upload to Facebook or Instagram handle, the videos we watch on YouTube and even the text messages we send all contribute to the gigantic heap that is unstructured data.
- Examples of unstructured data include text, video, audio, mobile activity, social media activity, satellite imagery, surveillance imagery etc.

- The Unstructured data is further divided into:
- Captured data:
 - It is the data based on the user's behavior.
 - The best example to understand it is GPS via smartphones which help the user each and every moment and provides a real-time output.
- User-generated data:
 - It is the kind of unstructured data where the user itself will put data on the internet every movement.
 - For example, Tweets and Retweets, Likes, Shares, Comments, on YouTube, Facebook, etc.
- Tools generates unstructured data:
 - Hadoop
 - HBase
 - Hive
 - Pip
 - MapR
 - Cloudera

Semi-Structured

- Semi Structured data is information that does not reside in a RDBMS.
- Information that is not in the traditional database format as structured data, but contains some organizational properties which make it easier to process, are included in semi-structured data.
- It may organized in tree pattern which is easier to analyze in some cases.
- Examples of semi structured data might include XML documents and NoSQL databases. Personal data stored in an XML file
 - `<rec><name>Angel Priya</name><sex>Male</sex></rec>`
 - `<rec><name>Babu Bhaiya</name><sex>Male</sex></rec>`
 - `<rec><name>Babita Ji</name><sex>Female</sex></rec>`
 - `<rec><name>Jethalal Tapu Ke Papa
Gada</name><sex>Male</sex></rec>`
 - `<rec><name>Dhinchak Pooja</name><sex>Female</sex></rec>`

Characteristics of each data type

Characteristics of each data type

Structured Data → Quantitative

- ✓ Numerical analysis
- ✓ Statistical calculations

Numbers
Dates
Predefined categories



Unstructured Data → Qualitative

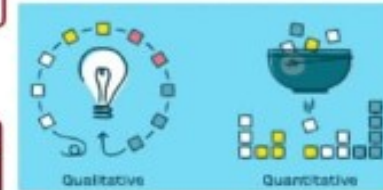
- ✓ Natural Language Processing
- ✓ Sentiment Analysis

Text
Images
Videos

Semi-structured Data → Quantitative + Qualitative

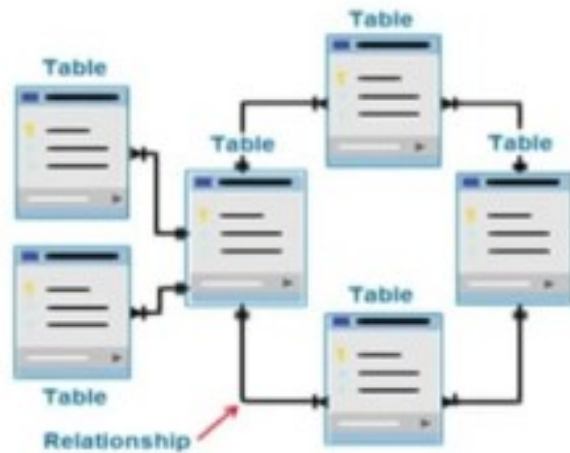
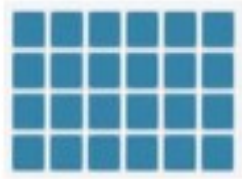
- ✓ Numerical values
- ✓ Free-text descriptions

Numbers
Descriptive
content



Where the data is stored and what are the different tools to work with each data type ?

Structured Data



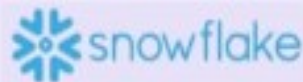
Organize and query data
- Predefined schema



Amazon
Redshift



Google BigQuery



Analytical processing

Complex queries



Structured data analysis

Characteristics of each data type

1. Unstructured Data

This includes things like:

- **Images**
- **Videos**
- **Documents**

These types of data don't follow a strict format (like a table or spreadsheet), so we call them **unstructured**.

 **Where are they stored?**

They are stored in **cloud services** like:

- **Amazon S3**
- **Google Cloud Storage**

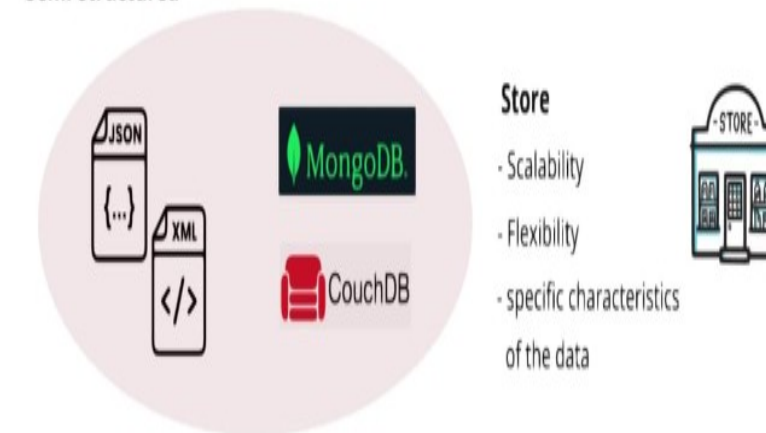
These services offer:

- **Scalable storage** (can grow as you need)
- **Durable storage** (safe and long-lasting)

Unstructured



Semi-structured



2. Semi-structured Data

This includes formats like:

- **JSON** (used in web APIs, configs, etc.)
- **XML** (used for documents, configs, etc.)

These are not as messy as unstructured data, but also not as strict as fully structured data like tables — so we call them **semi-structured**.

Where are they stored?

They are stored in databases like:

- **MongoDB**
- **CouchDB**

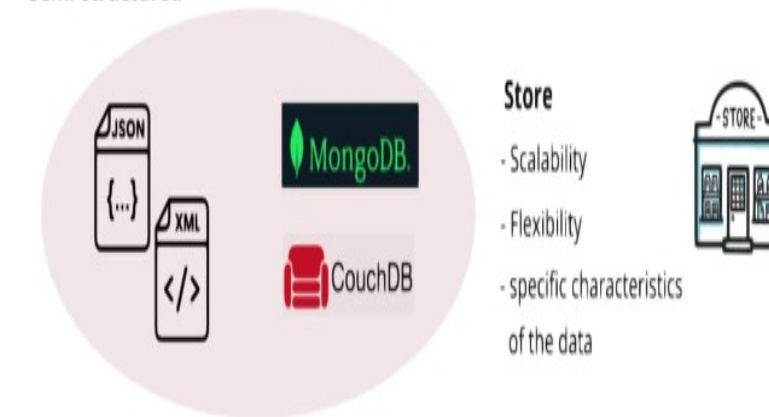
These databases offer:

- **Scalability** (can handle growing data)
- **Flexibility** (you don't need fixed table structures)
- Support for the **specific features of the data** (like nested values in JSON)

Unstructured



Semi-structured



Data Types and their different use cases:

Structured Data

Structured data is organized and easily searchable (e.g., data in rows and columns like in databases). Two common use cases are:

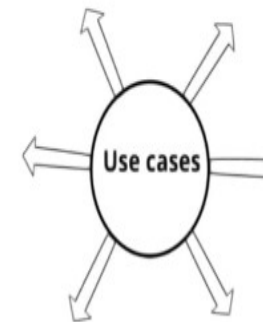
1. Customer Relationship Management (CRM)

1. **Purpose:** To understand customer behavior and patterns.
2. **Benefit:** Helps in **targeted marketing** and **increased customer engagement** by using structured information like purchase history, demographics, etc.

2. Supply-Chain Management

1. **Purpose:** To gain **real-time visibility** into inventory levels.
2. **Benefit:** Helps in **demand forecasting** and improves **operational efficiency** by analyzing structured data like stock quantities, delivery schedules, and supplier information.

Data types and their different use cases:



Structured data

Customer Relationship Management (CRM)

- Reveal customer behaviour and patterns for targeted marketing and increased engagement.

Supply-chain management

- Providing real-time visibility into inventory levels, facilitating demand forecasting, and enhancing overall operational efficiency.

Unstructured Data

Predictive Data Analytics: Alerts businesses about significant market shifts, enabling proactive planning and timely adjustments.

Fraud Detection in Finance: Analyzing unconventional patterns and behaviours to safeguard against financial irregularities.

Data Types and their different use cases:

Unstructured Data

Unstructured data includes formats like text, images, videos, and social media posts which don't follow a strict structure. Two key use cases are:

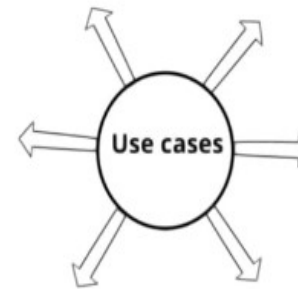
1. Predictive Data Analytics

1. **Purpose:** To alert businesses about major **market shifts**.
2. **Benefit:** Allows for **proactive planning** and **timely adjustments** using unstructured data sources like news articles, social media sentiment, etc.

2. Fraud Detection in Finance

1. **Purpose:** To detect **unconventional patterns and behaviors**.
2. **Benefit:** Helps **safeguard against financial fraud** by analyzing emails, transaction logs, voice recordings, etc.

Data types and their different use cases:



Structured data

Customer Relationship Management (CRM)

- Reveal customer behaviour and patterns for targeted marketing and increased engagement.

Supply-chain management

- Providing real-time visibility into inventory levels, facilitating demand forecasting, and enhancing overall operational efficiency.

Unstructured Data

Predictive Data Analytics: Alerts businesses about significant market shifts, enabling proactive planning and timely adjustments.

Fraud Detection in Finance: Analyzing unconventional patterns and behaviours to safeguard against financial irregularities.

Semi-structured data:

Customer Feedback Processing:

- Extracting valuable information to improve products, services, and overall customer satisfaction.

Log Files Analysis for IT Operations:

- Analyzed to detect anomalies, troubleshoot issues, and optimize system performance for enhanced reliability.

BIG DATA APPLICATION AREAS



Banking
& Securities



Communication,
Media & Entertainment



Healthcare



Manufacturing &
Natural Resources



Transportation



Education



Retail & Wholesale



Insurance

Healthcare & Public Health Industry

- Big Data has already started to create a huge difference in the healthcare sector.
- With the help of predictive analytics, medical professionals and HCPs are now able to provide personalized healthcare services to individual patients.
- Like entire DNA strings can be decoded in minutes.
- Apart from that, fitness wearables, telemedicine, remote monitoring—all powered by Big Data and AI are helping change lives for the better.

Academic

- Big Data is also helping enhance education today.
- Education is no more limited to the physical bounds of the classroom there are numerous online educational courses to learn from.
- Academic institutions are investing in digital courses powered by Big Data technologies to aid the all- round development of budding learners.

Banking

- The banking sector relies on Big Data for fraud detection.
- Big Data tools can efficiently detect fraudulent acts in real-time such as misuse of credit/debit cards, archival of inspection tracks, faulty alteration in customer stats, etc.

Manufacturing

- According to TCS Global Trend Study, the most significant benefit of Big Data in manufacturing is improving the supply strategies and product quality.
- In the manufacturing sector, Big data helps create a transparent infrastructure, thereby, predicting uncertainties and incompetence that can affect the business adversely.

IT

- One of the largest users of Big Data, IT companies around the world are using Big Data to optimize their functioning, enhance employee productivity, and minimize risks in business operations.
- By combining Big Data technologies with ML and AI, the IT sector is continually powering innovation to find solutions even for the most complex of problems.

Sports

- All sports popularly performing all analysis in big data analytics.
- The Cricket and football matches you will observe many predictions which found correct
- Example is IBM SlamTracker tool for tennis tournaments
- Also video analytics track the performance of every player in a cricket game, and sensor technology in sports equipment such as basketball allows us to get feedback even using smartphones.
- Many elite sports teams also track athletes outside of the sporting environment using smart technology to track nutrition and sleep, as well as social media conversations.

Science and Research

- Science and research is currently being transformed by the new possibilities big data.
- Experiments involve testing lot of possibilities of test cases and generate huge amounts of data.
- The many advanced lab uses computing powers of thousands of computers distributed across many data centers worldwide to analyze the data,
- It will help to leveraged performance areas of science and research.

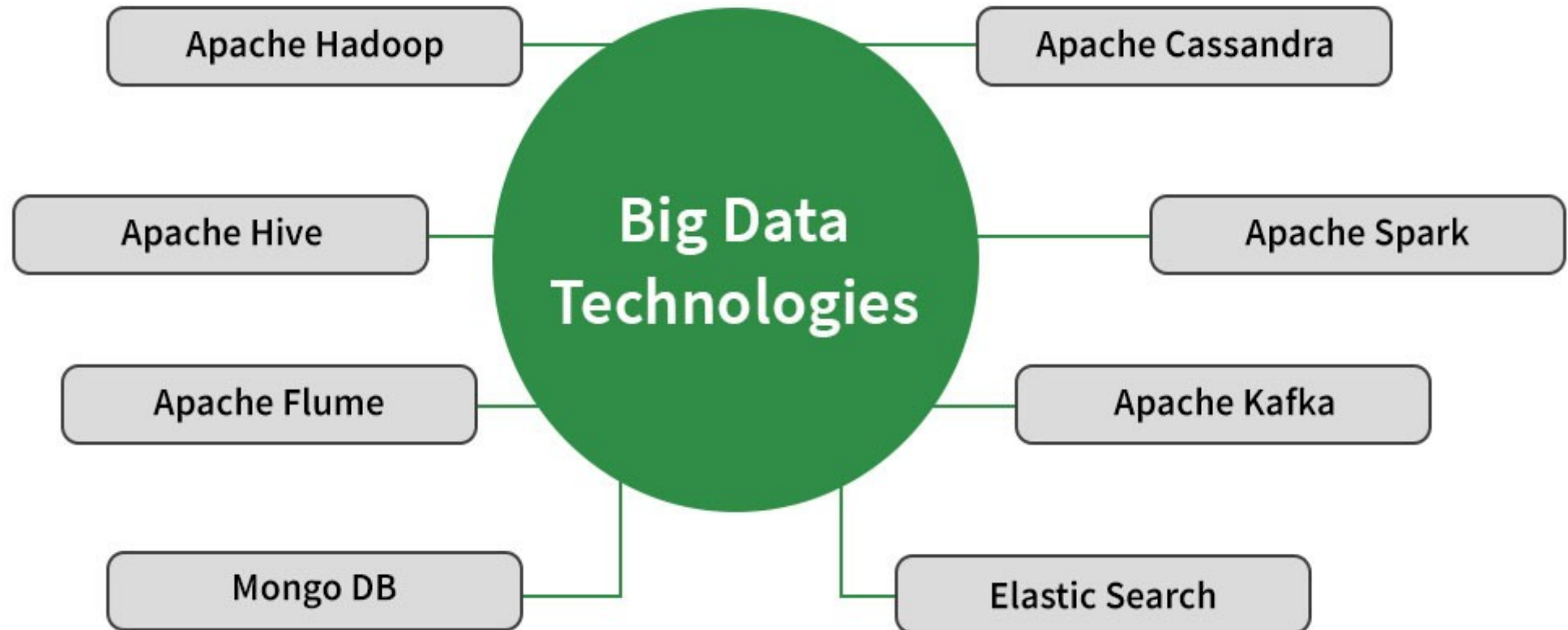
Security Enforcement

- Big data is applied for improving national security enforcement.
- The National Security Agency (NSA) in the U.S. uses big data analytics to foil terrorist plots.
- The big data techniques are used to detect and prevent cyber attacks.
- Police forces use big data tools to catch criminals and even predict criminal activity and credit card companies use big data use it to detect fraudulent transactions.

Financial Trading

- Automated Trading and High-Frequency Trading (HFT) is a new area where big data can play a role.
- The big data algorithms can be used to make trading decisions.
- Today, the majority of equity trading now takes place via data algorithms that increasingly take into account signals from social media networks and news websites to make buy and sell decisions split seconds

Popular Big Data technologies



Lifecycle of BDA

1. Business Case Evaluation

- Understand the goal or problem you're trying to solve.
- Example: A company wants to know why sales dropped last quarter.

2. Identification of Data

- Find out what data is needed and where it can be found.
- Example: Sales records, customer feedback, website data, etc.

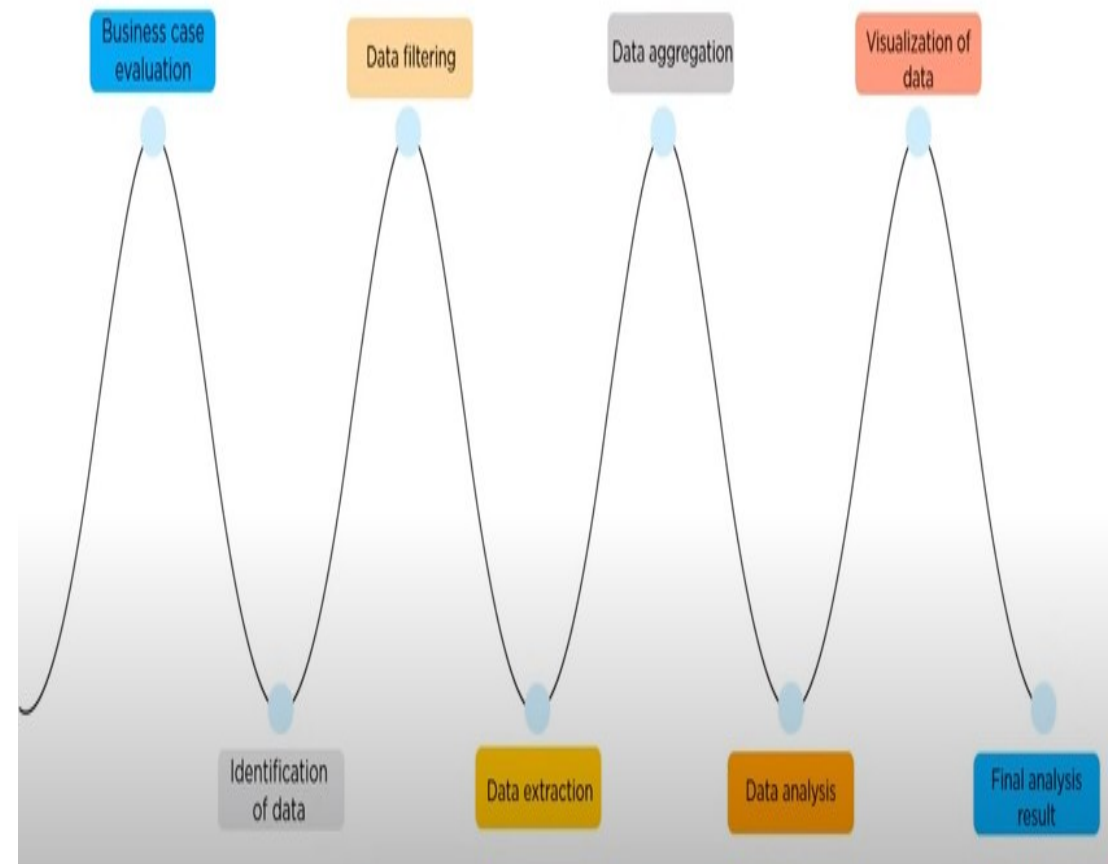
3. Data Filtering

- Remove irrelevant or unnecessary data.
- Only keep data that helps answer the business question.

4. Data Extraction

- Collect the data from various sources like databases, cloud, sensors, or social media.
- Example: Download sales data from the company's server.

Lifecycle of big data analytics



Lifecycle of BDA

5. Data Aggregation

- Combine data from different sources into one place.
- This makes it easier to analyze everything together.

6. Data Analysis

- Apply methods like statistics, machine learning, or data mining to find patterns and gain insights.
- Example: Analyzing which products sell best in certain seasons.

7. Visualization of Data

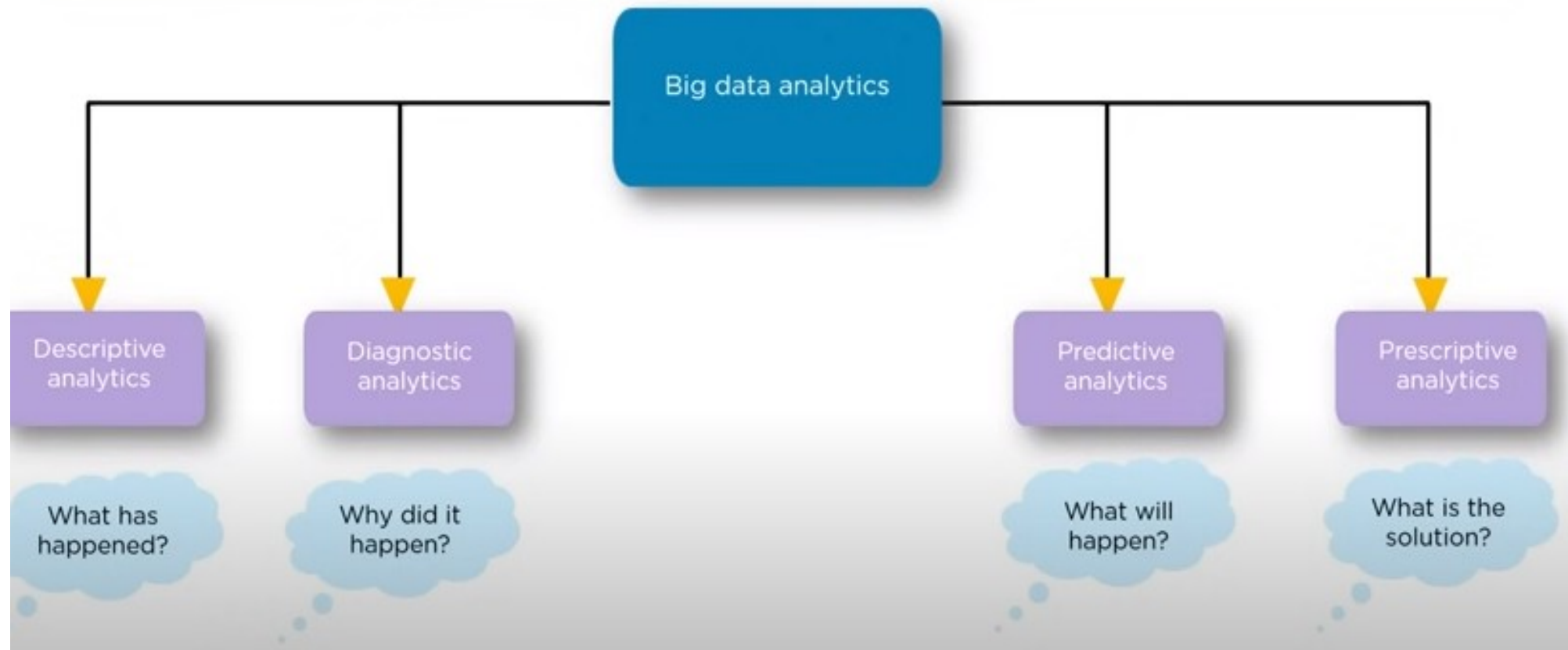
- Show the analysis results using graphs, charts, and dashboards.
- Makes it easier to understand and explain findings to others.

8. Final Analysis Result

- Make decisions or recommendations based on the insights.
- Example: Increase advertising for high-demand products or adjust pricing.

Types of Big data Analytics

Types of big data analytics



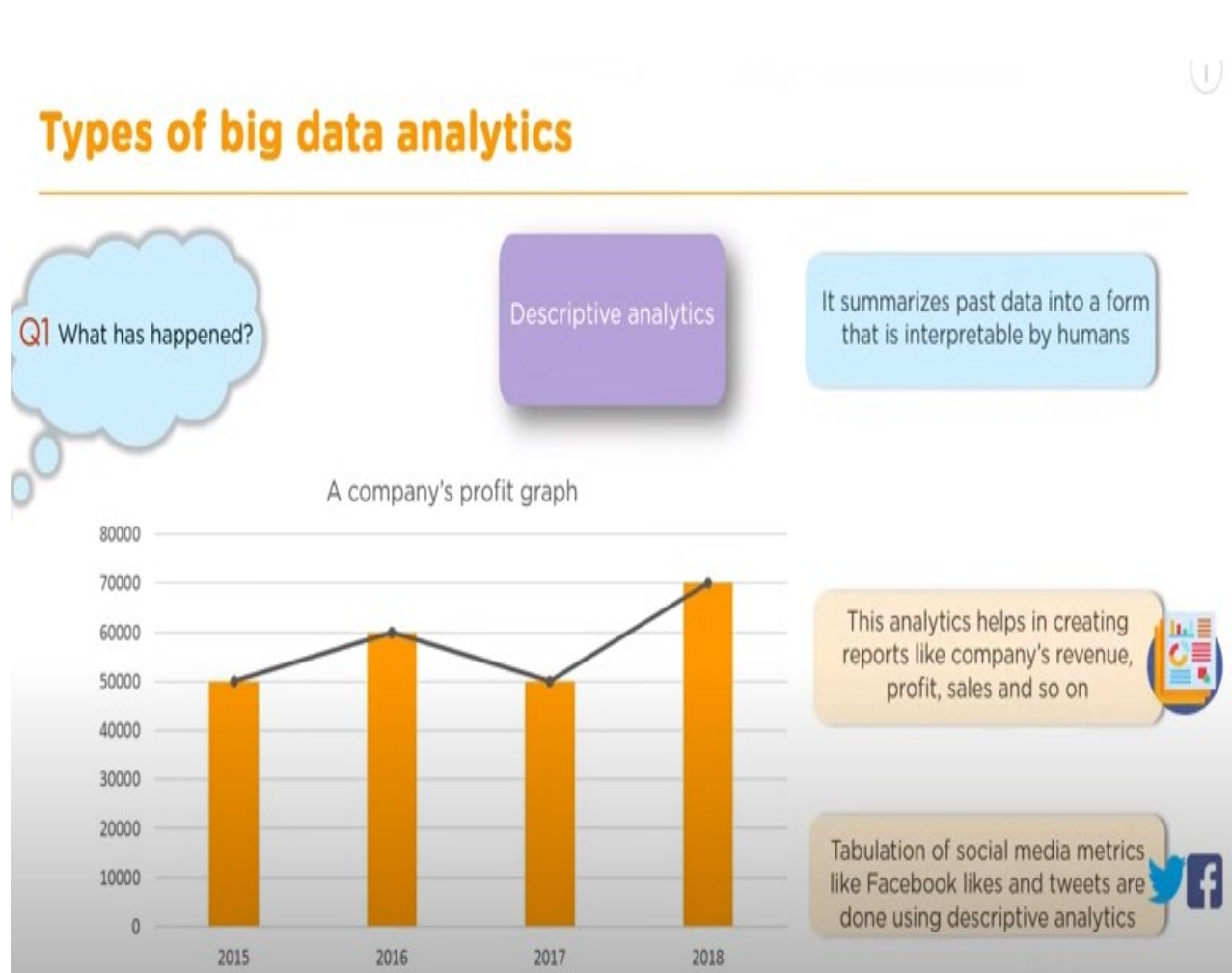
Descriptive Analytics

1. Descriptive Analytics – “What has happened?”

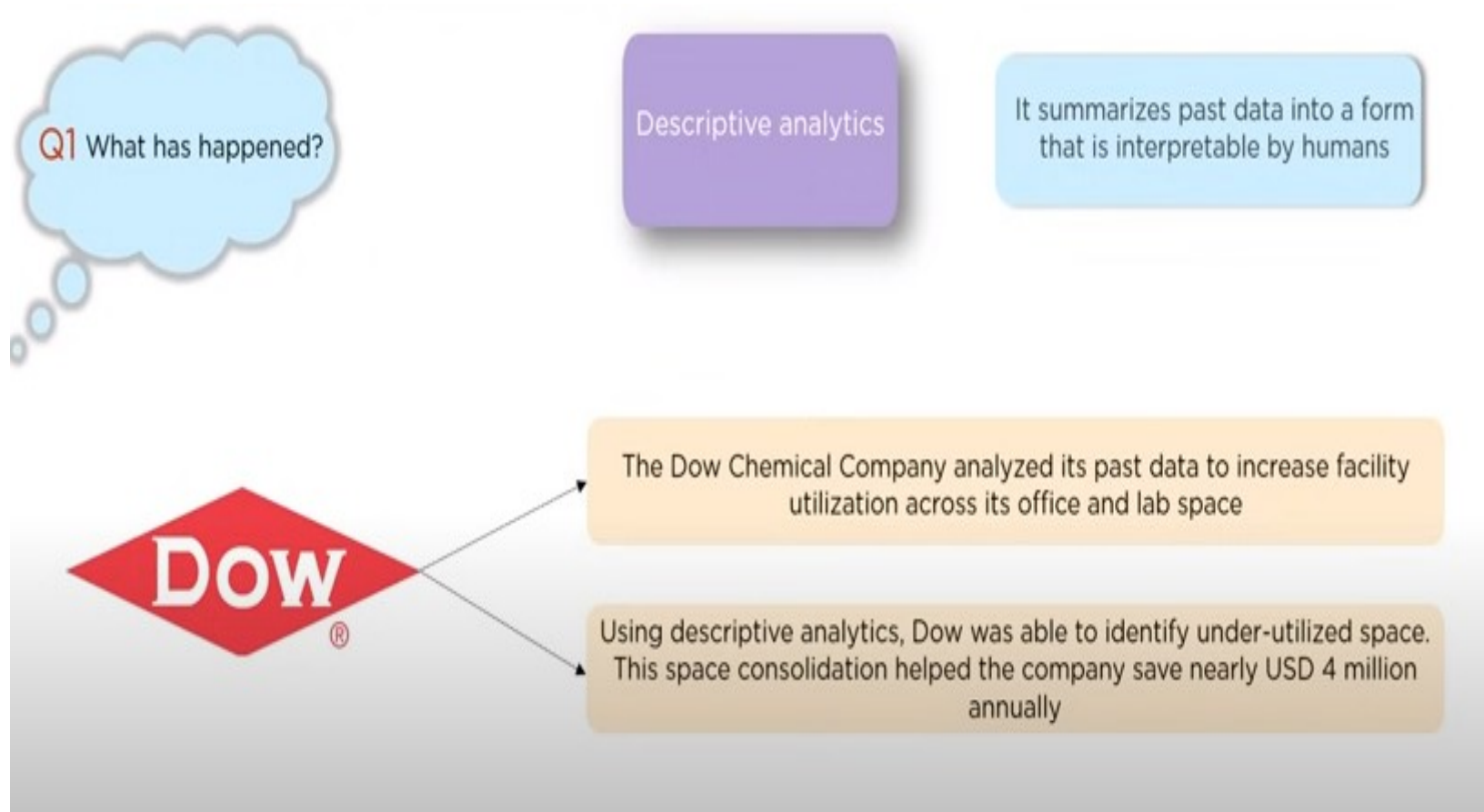
Goal: To understand past events.

Example: A company looks at last month's sales report to see which product sold the most.

Use: Helpful for reporting and summarizing what already occurred.



Use cases



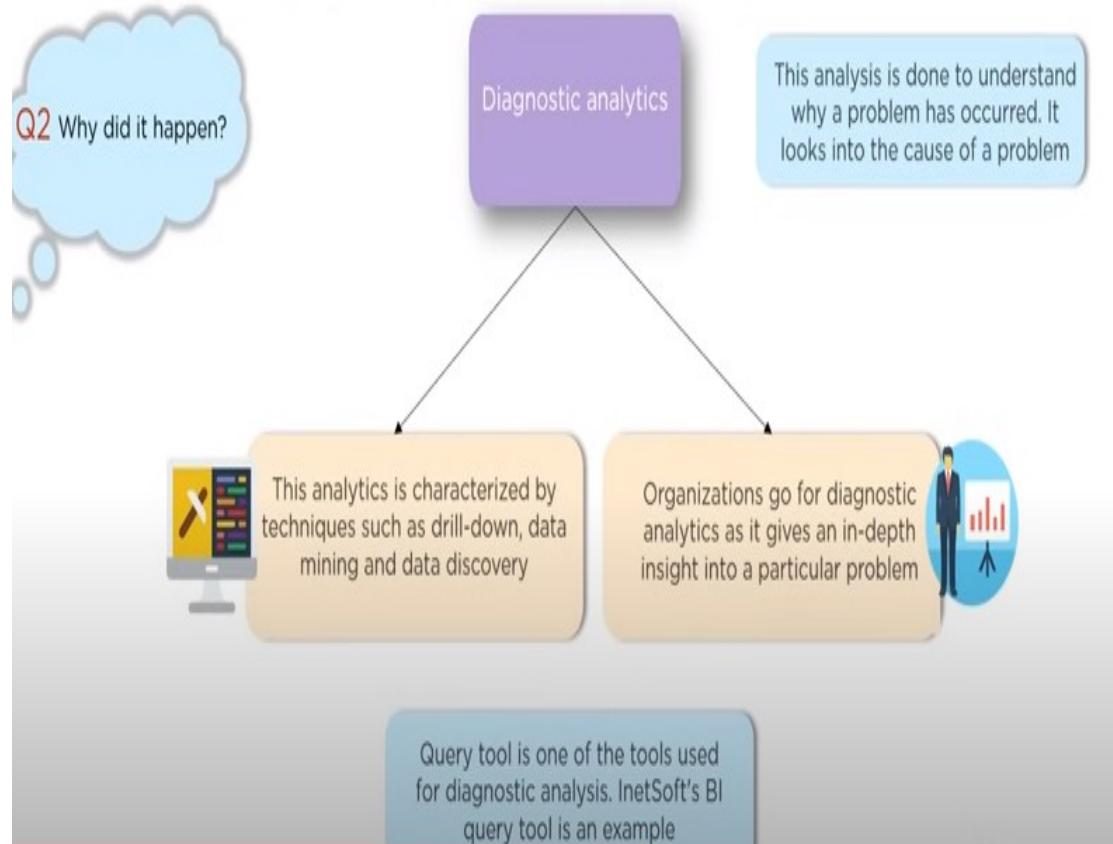
2. Diagnostic Analytics or behavioral analytics

2. Diagnostic Analytics – “Why did it happen?”

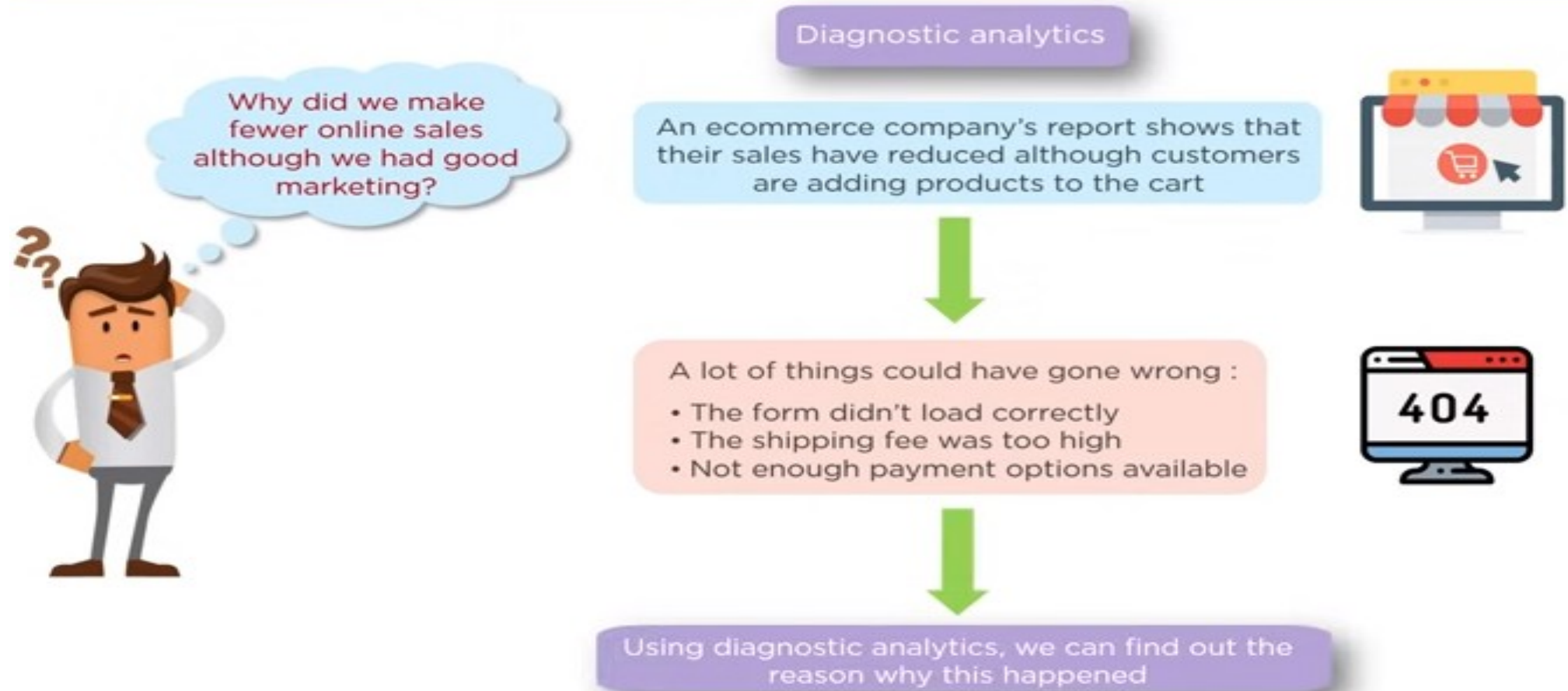
Goal: To figure out the reasons behind past outcomes.

Example: If sales dropped, this analysis helps discover the cause—like a marketing issue or product shortage.

Use: Useful for understanding root causes of successes or failures.



Use cases



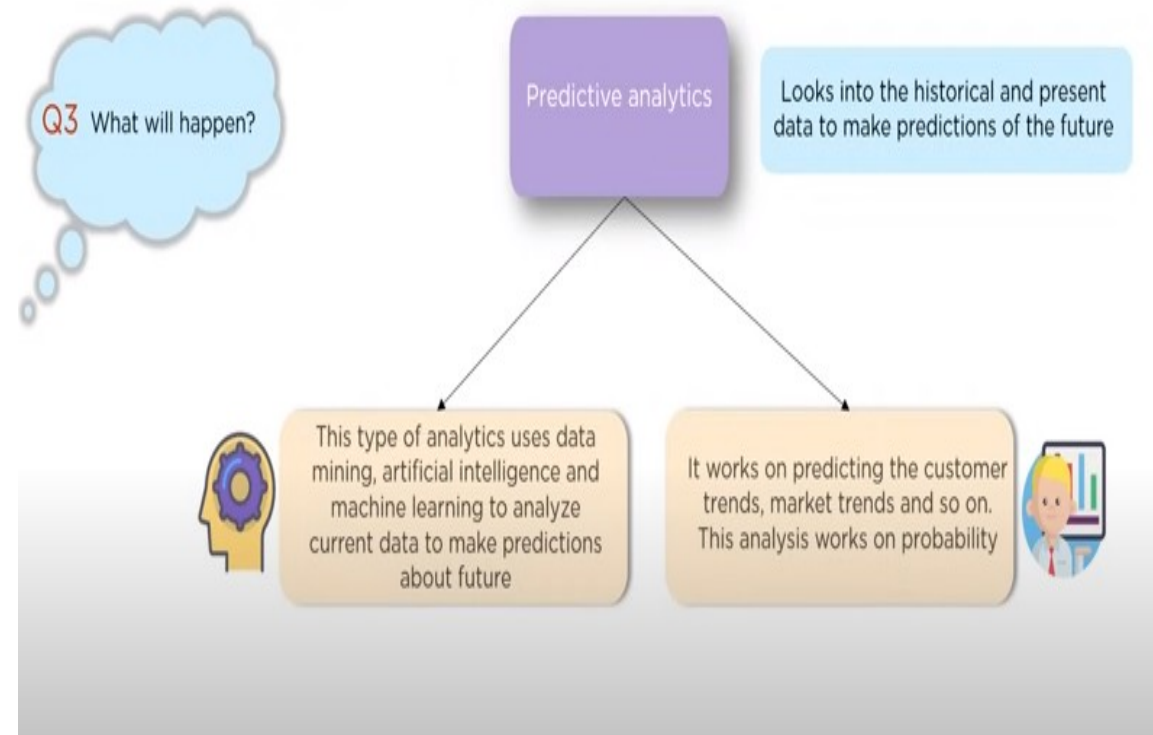
3. Predictive Analytics

◆ 3. Predictive Analytics – “What will happen?”

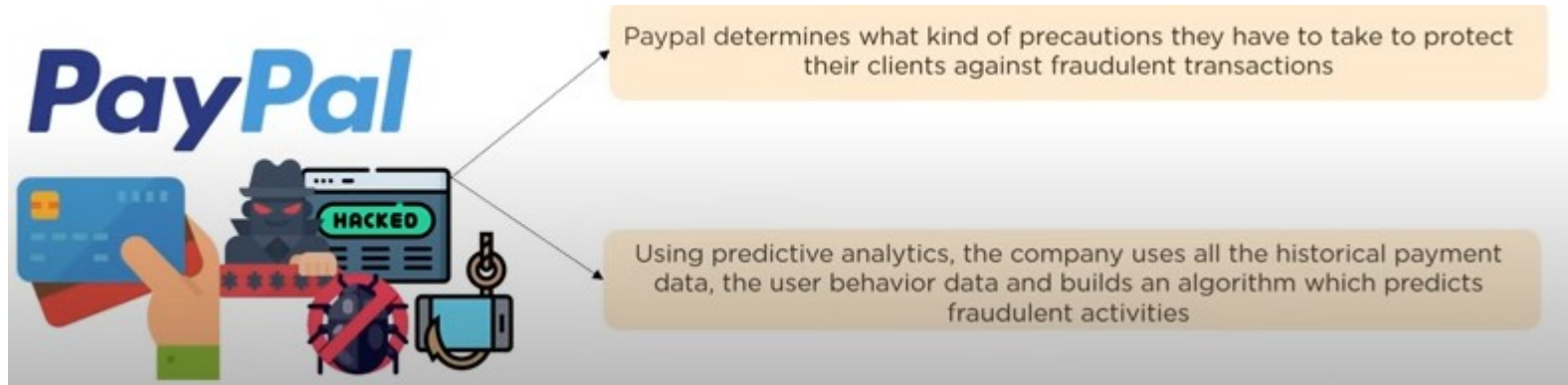
Goal: To forecast future outcomes based on data trends.

Example: A store predicts how much ice cream it will sell next summer based on past trends and weather patterns.

Use: Helps with planning and anticipating events.



Use cases



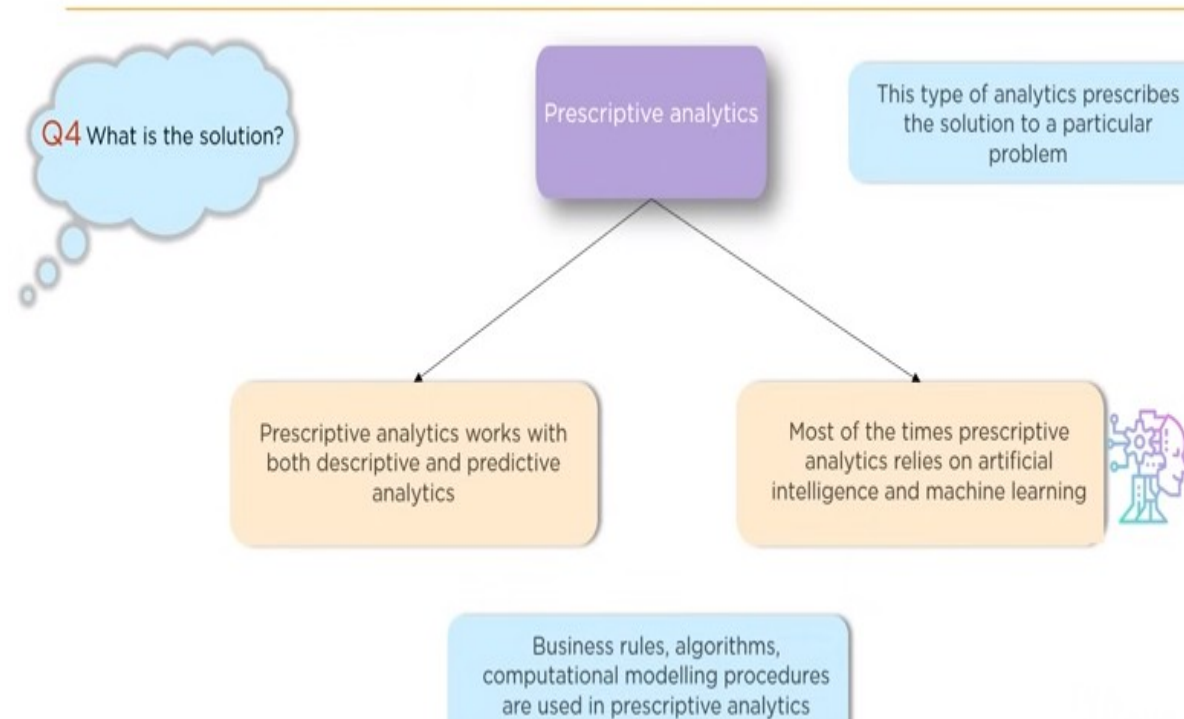
Prescriptive Analytics

These types of analytics are based on rules and recommendations and thus prescribe an analytical path. The analysis is generally based on the question, '**what actions should be taken?**' Google's self-driving car is an example of prescriptive analysis.

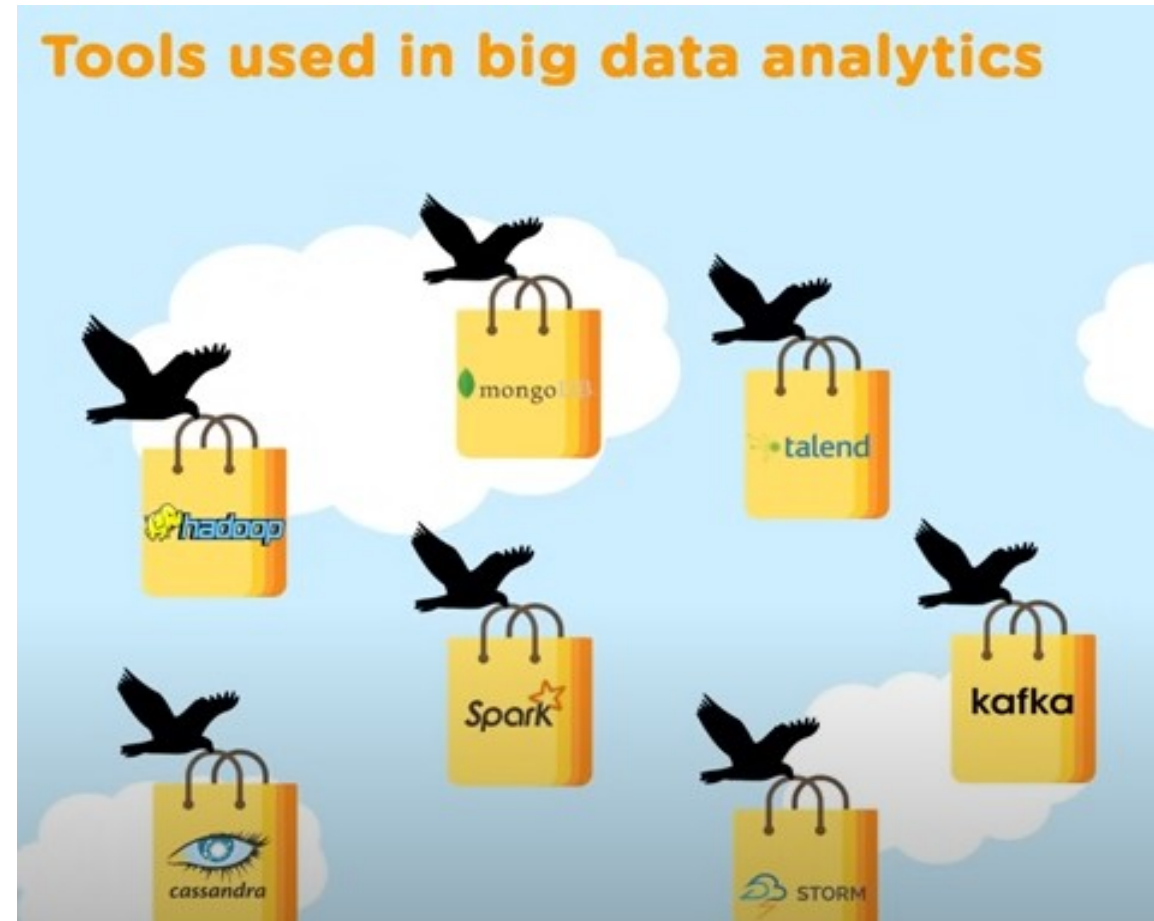
◆ 4. Prescriptive Analytics – “What is the solution?”

Goal: To recommend actions or decisions.

Example: Suggests increasing inventory or launching a sale to meet expected demand.



Tools used in BDA



Main Components of Big Data

- **1. Ingestion**

- Ingestion refers to the process of gathering and preparing the data. You'd use the ETL (extract, transform, and load) process to prepare your data. In this phase, you have to identify your data sources, determine whether you'll gather the data in batches or stream it, and prepare it through cleansing, massaging, and organization. You perform the extract process in gathering the data and the transformation process in optimizing it.

- **2. Storage**

- Once you have gathered the necessary data, you'd need to store it. Here, you'll perform the final step of the ETL, the load process. You'd store your data in a data warehouse or a data lake, depending on your requirements. This is why it's crucial to understand your organization's goals while performing any big data process.

- **3. Analysis**

- In this phase of your big data process, you'd analyze the data to generate valuable insights for your organization. There are four kinds of big data analytics: prescriptive, predictive, descriptive, and diagnostic. You'd use artificial intelligence and machine learning algorithms in this phase to analyze the data.

- **4. Consumption**

- This is the final phase of a big data process. Once you have analyzed the data and have found the insights, you have to share them with others. Here, you'd have to utilize data visualization and data storytelling to share your insights effectively with a non-technical audience such as stakeholders and project managers.

CHALLENGES IN HANDLING BIG DATA

The Bottleneck is in technology

- New architecture, algorithms, techniques are needed

Also, in technical skills

- Experts in using the new technology and dealing with big data
- Need For Synchronization Across Disparate Data Sources
- Getting Meaningful Insights Through The Use Of Big Data Analytics
- Getting Voluminous Data Into The Big Data Platform
- Data Storage And Quality
- Security And Privacy Of Data

BIG DATA APPLICATIONS

- Natural Language processing
- Human behaviour monitoring
- Geographic information systems
- Bio-informatics
- Medical science
- Weather forecasting
- Cloud control systems
- Multimedia
- Body sensor Networks

BIG DATA TECHNIQUES

The processing of Big data can be done by two processes:

1. Batch Processing
2. Non batch processing

Batch Processing: In batch processing, data is collected over a period of time processed and then produces an output. It uses MapReduce framework.

Non batch processing : Involves continual input of data and processed in small period of time. Real time big data analysis or stream processing.

CASE STUDY..

- In the field of infectious disease (The three Academies work together as the National Academies of Sciences, Engineering, and Medicine to provide independent, objective analysis and advice to the nation and conduct other activities to solve complex problems and inform public policy decisions. The National Academies also encourage education and research, recognize outstanding contributions to knowledge, and increase public understanding in matters of science, engineering, and medicine.)

DATA ANALYTICS APPLICATIONS IN AROGYA SETU APP

1. The Aarogya Setu app starts by asking your mobile number to sign you up. It then sends you a privacy notice to inform you that the app will collect and use certain data.
2. It will then request access to your device location following which you need to switch your Bluetooth on for 120 seconds.
3. Your Bluetooth and GPS needs to be continuously switched on for the app to work. As it needs to keep collecting your data and analyse it to draw meaningful conclusions.
4. The app also requires additional data such as full name, gender, age, travel history of the last 30 days, and professional details.

5. Aarogya Setu also entails a self-assessment questionnaire wherein it asks:

Gender and age

cough, fever or difficult breathing symptoms

diabetes, hypertension, lung/ heart disease

International travel over the last 14 days.

Questions pertaining to recent interactions with COVID-positive individuals if you are a healthcare professional and have examined a COVID-positive case without protective gear.

6. The app will then leverage the collected data and use data analytics to put the person under low risk or high-risk category.

7. The app confirms that your data will only be shared with the Government of India and will never disclose your name/number to the public at any given time.

8. In order to draw conclusions, the app continuously collects your location data. It also keeps a record of all the places you came in contact with other app users.

9. The app suggests that the Government will only upload and use your personal information in “anonymized, aggregated datasets.” (Anonymised: removal of personally identifiable information from data sets). This is further utilised for COVID-19 management by generating reports, heat maps, and other statistical visualizations.

10. Aarogya Setu app continuously seeks personal information, location details, and health data. This raised data privacy concerns. What will be shared, with whom will it be shared, and what can be the potential purpose were the common questions.

All of this was mainly because the government was silent until now on the security practices of the application. A recent update, however, confirms that the app is now open source. The decision of making the source code public will aid experts to detect the vulnerabilities of the application. More and more people will be able to rely on the app as the developers can now spot issues with the code if any.

DATA ANALYTICS APPLICATIONS TO COMBAT COVID AROUND THE WORLD

1. QuadMed (Sussex) recently announced leveraging data analytics to learn more about COVID-19 risk factors. For instance, to determine the segment of people who are most likely to contract the coronavirus, scientists can study the links between individuals' age/health conditions and COVID-19.

2. The Italian health system aims to leverage data analytics to identify complications, allocate resources, and pluck clues from medical records.

It finally aims to personalise the services for each patient and predict the duration of their hospital stays. Scientists strive to utilise data analytics and enhance the effectiveness of decisions for patients.

How Bluetooth and Data analytics lead contact tracing for Covid-19:

<https://youtu.be/3qBh9lFqLs8>